Research on damping and energy recovery characteristics of a novel mechanical-electrical-hydraulic regenerative suspension system

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Abstract: The suspension vibration energy generated by tracked vehicle under high-speed off-road driving conditions is huge, and the existing regenerative suspension power density cannot meet its kilowatt-level energy recovery requirements. In this paper, a novel high-power density mechanical-electrical-hydraulic (MEH) regenerative suspension system is proposed based on the bidirectional energy management control strategy. The hydraulic motor-generator is used as the core component of hydraulic-electrical energy conversion, which realizes high power energy conversion in a limited space. Based on the energy transfer mechanism of MEH regenerative suspension, the multiphysics coupling simulation model of the suspension and the energy recovery module test device are built. And then the energy recovery effect of the MEH regenerative suspension system is analyzed by combining simulation and experiment. The test results show that the energy recovery module can effectively realize energy conversion and recovery, and the maximum hydraulic-electrical energy conversion efficiency can reach 40% under 5Ω external resistance. On this basis, it is further obtained through simulation that the MEH regenerative suspension system can realize the damping force adjustment under passive and semi-active working conditions, and its energy recovery power can reach the kilowatt level, which has an excellent energy recovery characteristic.

Keywords: Tracked vehicle, integrated regenerative suspension, energy conversion, energy recovery characteristic, damping characteristic

1. Introduction:

High-speed tracked vehicles are mostly undulating off-road roads, and the vibration energy generated under high-speed and heavy-load driving conditions is huge.

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The traditional passive suspension dissipates the vibration energy in the form of thermal energy, which not only causes energy waste but also attenuates the performance of the shock absorber seriously to the rise in temperature, which poses a huge challenge to the reliability of the suspension system [1]. If the suspension vibration energy can be efficiently recovered under different road conditions, it can not only improve the energy utilization efficiency of high-speed tracked vehicles, improve its comprehensive performance such as vehicle endurance, but also effectively reduce the temperature rise of the shock absorber, improve its working reliability and performance. Therefore, it is of great significance to carry out research on the regenerative suspension system for high-speed tracked vehicles.

The regenerative suspension system can realize suspension vibration energy recovery while improving the vehicle vibration damping performance. The recovered energy is converted into electrical energy through the energy recovery circuit for active suspension [2] or stored in the battery for other electrical equipment to improve the energy utilization rate [3]. Zhang et al. [4] analyzed and summarized the energy recovery potential of various vehicles. The study shows that vehicle mass and road surface roughness are important factors affecting the energy recovery potential of vehicles. Heavy-duty vehicles driving on off-road roads have higher energy recovery potential. Mohamed et al. [5-7] studied the energy recovery potential of passenger cars, buses, trucks, and off-road vehicles under different driving conditions. The simulation calculation shows that the total energy recovery potential of the heavy-duty semi-trailer truck is 2-4kW when driving at a speed of 20m/s on the C-class road. Peng [8] installed the regenerative suspension on a military wheeled support vehicle for a real vehicle test. The test results show that the regenerative power of the vehicle can reach more than 100 watts when driving on the C-class road. His research also shows that the energy recovery potential of tracked vehicles is 4.37 times that of wheeled vehicles, and the regenerative suspension is more suitable for off-road vehicles, military vehicles and heavy-duty vehicles.

The regenerative suspension system can be divided into linear-motor type [9], electromechanical type [10] and electro-hydraulic type [11] according to the different

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energy transfer mechanisms of the shock absorber. Bose has developed a linear motor type regenerative suspension [12], which can realize active control and suspension vibration energy recovery at the same time. The prototype was installed on the Lexus LS400 for real vehicle tests and the results show that its ride comfort and handling stability are significantly improved compared to the car with traditional passive suspension. But this suspension has not been commercialized due to its cost and weight. Zuo et al. [13] conducted a theoretical study and optimized the design of the linear motor regenerative suspension and the scaled prototype bench tests show that when the root mean square value of the suspension speed was 0.25-0.5m/s, the energy recovery power can reach 16-64W. Li et al. [14,15] designed a mechanical motion rectifier for a rack-pinion-based regenerative suspension, which can convert the oscillatory vibration into unidirectional rotation of the generator to improve the energy recovery efficiency. The developed prototype is installed in an SUV, and the test results show that the average energy recovery power of the suspension is 15.4W when the load resistance is 20Ω. Yu et al. [16] designed a ball screw based regenerative suspension with double overrunning clutches, which can realize motion rectification and improve energy recovery efficiency. The test data shows that this kind of suspension can effectively improve the ride comfort and energy efficiency of the vehicle. Maravandi et al. [17, 18] developed a six-link regenerative shock absorber, which converts linear motion into rotational motion through a link structure and drives the motor to generate electricity through a speed-increasing gear to achieve energy recovery. The bench test shows that reducing the resistance can improve the energy recovery efficiency, and the maximum efficiency is about 56%.

The electromechanical regenerative suspension has a low bearing capacity and is only suitable for passenger vehicles. The electro-hydraulic regenerative suspension can provide higher damping force and is more suitable for off-road vehicles, heavy trucks, military vehicles, and other vehicles with heavy weight and poor driving conditions [4]. Guo et al. [19-21] designed a hydraulic interconnected regenerative suspension for heavy-duty vehicles, using four check valves to form a hydraulic bridge to achieve hydraulic rectification, which can achieve an average energy recovery power of 220W.

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Zhang et al. [22,23] designed a hydraulic cylinder with its own rectification function, and based on this, a new type of electro-hydraulic regenerative suspension was studied. The bench test for off-road vehicles shows that the damping characteristic curve of the suspension system is smooth and full, with good damping characteristics, and the average energy recovery power is 110.6W, which can increase the fuel efficiency by 5%. Qin et al. [24,25] conducted research on a new type of hydraulic interconnected regenerative suspension for road and rail vehicles. The bench test results show that the suspension can improve the ride comfort and handling stability of the vehicle at the same time. And under rough gravel road driving conditions, the average recovered power reaches 364.7W.

The above-mentioned regenerative suspension has a limited power density and is only suitable for wheeled vehicles, and cannot meet the kilowatt-level energy recovery power of high-speed tracked vehicles. Volvo has developed a planetary gear type regenerative suspension system for high-speed tracked vehicles [26,27]. The planetary gear transmission mechanism is used to achieve acceleration and torque reduction in passive conditions to improve energy recovery efficiency. However, the bench test of the China Northern Vehicle Research Institute has proved that the alternating inertial dynamic load generated by the frequent commutation of the electromechanical suspension system is extremely large, which greatly reduces the reliability and fatigue life of the suspension [28].

Aiming at the above problems, a novel MEH regenerative suspension system with high reliability, low hysteresis and high-power density suitable for high-speed tracked vehicles is proposed. In the following content, the working principle and energy transfer path of the MEH regenerative suspension system are demonstrated first. Then carry out simulation and experimental research on its energy recovery characteristics to verify its energy recovery effect.

2. Suspension operating mechanism and modeling

The MEH regenerative suspension system can convert the vibration energy generated by the high-speed tracked vehicle driving on uneven roads into electrical

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energy through the energy recovery module, so as to improve the energy utilization rate.

2.1 Design and working principle

The energy recovery principle of the MEH regenerative suspension system is shown in Fig. 1. When the high-speed tracked vehicle is running on uneven roads, the connecting rod converts the bouncing up and down of the bogie wheel into the reciprocating swing of the pull arm, which transfers the vibration energy to the suspension actuator. The MEH suspension actuator is shown in Fig. 2, it adopts an integral structure with concentrated functions, and its volume is only 0.019m^3, which has a high-power density. Inside the suspension actuator, the hydraulic oil in the vane damper is hydraulically rectified through the hydraulic commutator composed of four check valves arranged on the diaphragm, and then the hydraulic oil is unidirectionally input to the hydraulic motor through the valve plate, drives the generator to rotate and generate electricity. The accumulator arranged on the high-pressure side can effectively buffer the hydraulic shock and improve the stability of the system.

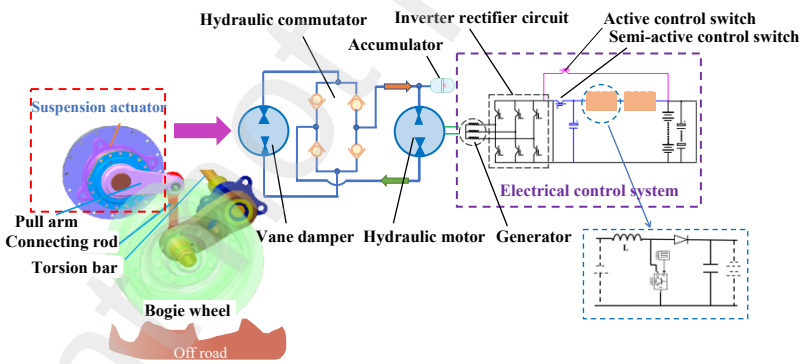

Fig. 1. Working principle of the MEH regenerative suspension system.

The four-quadrant hydraulic motor-generator is the core component of the hydraulic-electrical energy conversion of the MEH regenerative suspension system, which is composed of an internal meshing pump/motor and a permanent magnet motor. The structure is shown in Fig. 3. The hydraulic motor is wrapped inside the generator rotor and integrated with the generator coaxially. It integrates self-lubrication, cooling, flow distribution, and energy conversion. It has a high-power density and can realize high-power energy conversion and transmission in a limited space. The four-quadrant

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hydraulic motor-generator has the functions of motor, generator, hydraulic motor and hydraulic pump at the same time, which provides the basis of key components for realizing bidirectional energy transfer and passive/semi-active/active layered control of the MEH regenerative suspension system. This paper mainly studies the passive/semi-active energy recovery conditions, in which the hydraulic motor-generator acts as a hydraulic motor and a generator.

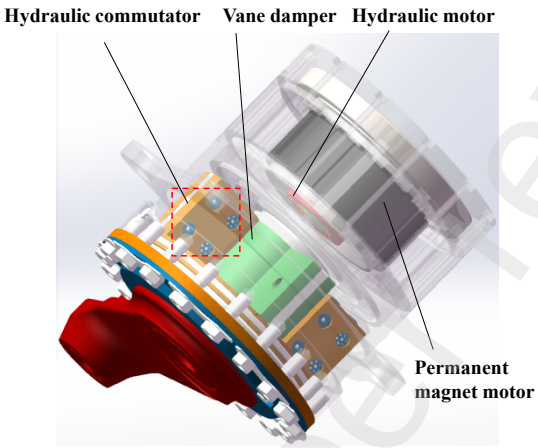

Fig. 2. Suspension actuator

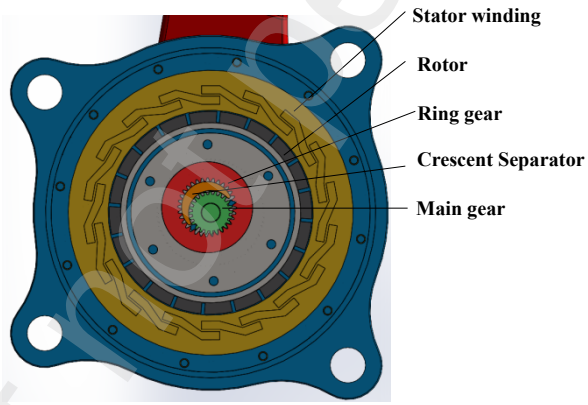

Fig. 3. Hydraulic motor-generator

The unstable three-phase alternating current generated by the permanent magnet motor is converted into a stable direct current through the inverter rectifier circuit, the buck-boost circuit and the filter circuit, which charges the vehicle battery and completes the energy recovery. The inverter rectifier circuit adopts a PWM rectifier, which can realize the bidirectional flow of energy, and has the advantages of high-power factor and fast dynamic response. Buck-Boost circuit can maintain the stability of battery charging voltage, and improve charging efficiency and battery life. The circuit schematic diagram is shown in Fig. 1. When the IGBT switch is turned on, the rectified

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direct current charges the energy storage inductor L, and the inductor current increases; when the IGBT switch is turned off, the energy storage inductor supplies power to the battery through the freewheeling diode, and the inductor current decreases.

If the minimum value of the inductor current is greater than zero, the system reaches the steady state of the inductance volt-second conservation as shown in equation (1) [29]:

$$U_i \cdot T_{\text{on}} = U_o \cdot T_{off} \tag{1}$$

where U_i and U_o are the input voltage and output voltage of the Buck-Boost circuit respectively; T_{on} and T_{off} are the on and off times of the IGBT respectively.

Equation (1) can be rewritten as:

$$U_i D T_i = U_o (1 - D) T_i \tag{2}$$

where T_i is the period of the IGBT switch control signal; D is the duty cycle of the IGBT switch control signal, $D = T_{\text{on}}/T_i$.

Then the relationship between the output voltage and the input voltage can be calculated as:

$$U_o = \frac{D}{1 - D} \cdot U_i \tag{3}$$

It can be known from equation (3) that when the duty ratio is less than 0.5, the buck output can be achieved, and when the duty ratio is greater than 0.5, the boost output can be achieved.

The electrical control system can not only realize vibration energy recovery, but also realize passive/semi-active layered control through the energy reverse management control strategy. On the premise of meeting the vibration reduction requirements, the energy can be reasonably used to achieve the highest energy recovery efficiency.

2.2 Energy transfer path

During the energy transmission process of the MEH regenerative suspension system, the vibration energy undergoes the conversion from hydraulic energy to

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mechanical energy and finally to electrical energy. During the transfer and conversion process, the vibration energy will be lost in various forms and cannot be completely converted into electrical energy and recovered. The transfer path is shown in Fig. 4.

The suspension vibration energy generated by the road excitation is first transmitted to the vane damper through the connecting rod and the pull arm to realize the conversion of vibration energy to hydraulic energy. Due to its own structural characteristics, the vane damper has a large leakage during operation, and there is a large volume loss during the transmission of hydraulic energy to the hydraulic commutator, resulting in hydraulic energy loss. When the hydraulic oil flows through the hydraulic commutator, due to the opening pressure of the check valves, there is a certain pressure loss during the high-frequency reciprocating flow, resulting in hydraulic energy loss. The hydraulic oil enters the hydraulic motor and drives the motor to rotate to realize the conversion of hydraulic energy to mechanical energy. In this process, energy loss will be generated due to the frictional flow resistance of the hydraulic motor. The mechanical energy output by the hydraulic motor drives the generator to generate electricity to complete the conversion of mechanical energy to electrical energy. During this process, the parasitic energy loss of the generator will consume part of the electrical energy. The energy recovery circuit recovers the electrical energy output by the generator and stores it in the battery to complete the vibration energy recovery, but in the process, part of the energy will be consumed due to circuit loss.

It can be seen from the above analysis that the vibration energy undergoes the conversion and transmission process of hydraulic energy to mechanical energy and then to electrical energy in the MEH regenerative suspension system, and there are many energy losses in this process. Therefore, optimizing the structure and reducing the loss in the energy conversion and transmission process is of great significance to improve the energy recovery efficiency of the MEH regenerative suspension system.

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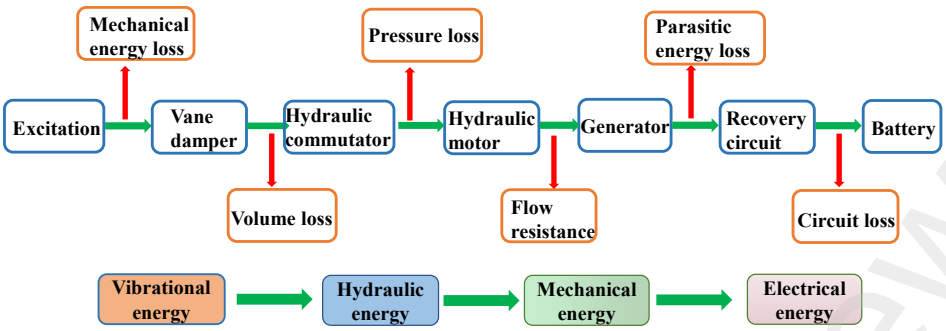

Fig.4. Energy transfer path of the MEH regenerative suspension system

2.3 Mathematical model

2.3.1 Analysis of hydraulic circuit

The MEH regenerative suspension system adopts the vane damper as the vibration damping element. It is known from the working principle of the vane damper that whether the vane damper rotates clockwise or counterclockwise, the total flow in the working chamber is equal to the volume swept by the vane per unit of time. The flow can be calculated by:

$$Q_{vane} = 2\frac{v}{r_b}b\int_{D_n/2}^{D_w/2} x dx = \frac{b(D_w{}^2 - D_n{}^2)}{4r_b}v \tag{4}$$

where Q_{vane} is the flow rate of the vane damper; v is the tangential speed of the outer end of the pull arm; r_b is the length of the pull arm; b is the working height of the vane; D_w and D_n are the outer diameter and inner diameter of the vane damper respectively.

The vane damper is divided into two high-pressure chambers and two low-pressure chambers by the diaphragm and the vane. When the vane rotates, the hydraulic oil in the two high-pressure chambers is rectified by the two hydraulic commutators arranged on the diaphragm, and then flows into the high-pressure chamber of the hydraulic motor to push the hydraulic motor to rotate. The hydraulic oil discharged from the low-pressure chamber of the motor flows back to the low-pressure chamber of the vane damper through the hydraulic commutator. The hydraulic oil flow directions in the suspension actuator when the vane damper rotates clockwise and

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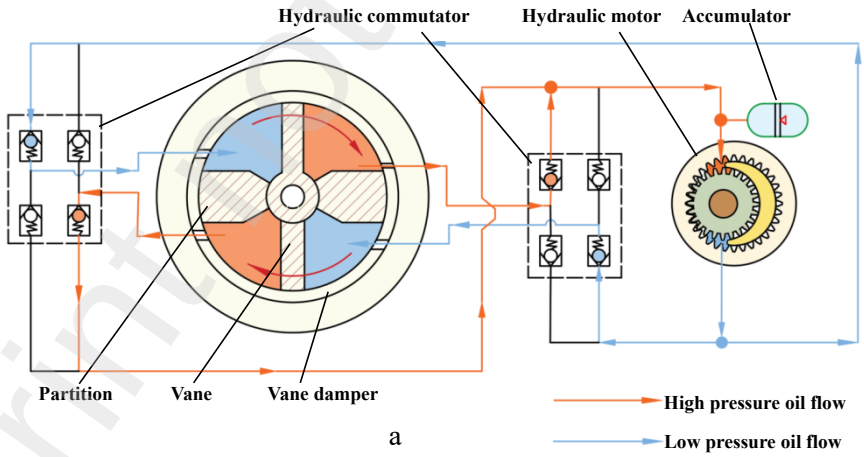
counterclockwise are shown by Fig. 5(a) and Fig. 5(b) respectively. It can be seen from the hydraulic circuit that no matter whether the vane damper rotates clockwise or counterclockwise, the hydraulic oil can always flow into the hydraulic motor from one direction, and drive the hydraulic motor to rotate in one direction to drive the generator to generate electricity, thereby improving the energy recovery efficiency. Under large amplitude conditions, when the hydraulic circuit pressure is high, the accumulator stores part of the hydraulic oil, and releases part of the hydraulic oil when the circuit pressure is insufficient to maintain the circuit stability. The hydraulic oil flowing into and out of the accumulator can be regarded as dynamic balance. The flow rate of hydraulic oil through each component is:

$$Q_v = Q_p = Q_m = \eta_{v1} Q_{vane} \tag{5}$$

where Q_v is the flow rate of the hydraulic commutator, which can be equivalent to the flow rate through one check valve; Q_p, Q_m and Q_{vane} are the flow rate of the hydraulic pipe, hydraulic motor and vane damper respectively; η_{v1} is the volumetric efficiency of the vane damper.



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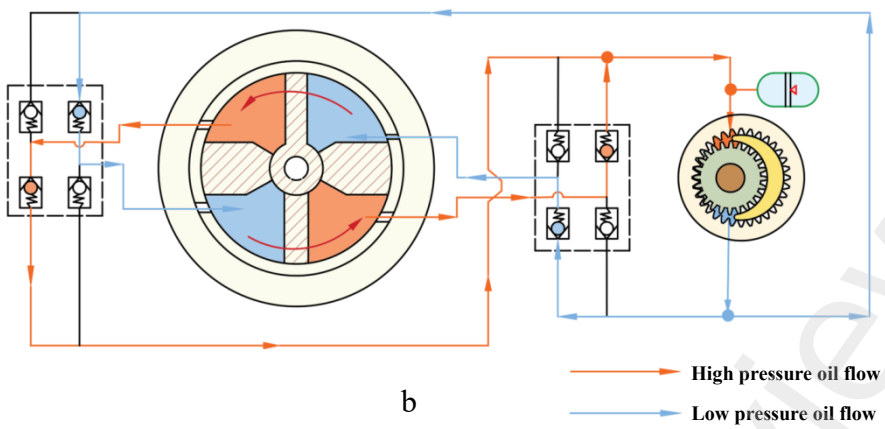

b

Fig.5. Hydraulic circuit of MEH regenerative suspension system. (a) clockwise, (b) counterclockwise.

2.3.2 Pressure drop of check valve

The check valve used in the hydraulic commutator is a tubular check valve. In order to reduce the pressure loss when the hydraulic oil flows through the hydraulic commutator, the check valve is set to a lower opening pressure. The orifice of the check valve is equivalent to a thin-walled hole, so the flow through the check valve can be calculated according to the thin-walled orifice formula:

$$Q_{vi} = C_q A_{ci} U_{vi} = C_q A_{ci} \sqrt{2\Delta P_{vi}/\rho} \tag{6}$$

where C_q is the flow coefficient of the hydraulic oil; ρ is the hydraulic oil density; A_{ci} , U_{vi} and ΔP_{vi} are the throttling area, flow rate of hydraulic oil and pressure drop of the check valve i respectively.

Therefore, the pressure drop of the check valve can be calculated as:

$$\Delta P_{vi} = \frac{\rho {Q_{vi}}^2}{2{C_q}^2{A_{ci}}^2} \tag{7}$$

2.3.3 Pressure drop of hydraulic pipeline

The vane damper and the hydraulic motor exchange hydraulic oil through the valve plate. The flow of hydraulic oil in the pipeline of the valve plate is in a laminar state. Ignoring the local pressure loss of the hydraulic pipeline, only considering the pressure loss along the way, the pressure drop of the pipeline can be calculated as:

$$\Delta P_p = \frac{128\mu l_p}{\pi {d_p}^4} Q_p \tag{8}$$

where ΔP_p is the pressure drop of the hydraulic pipeline; μ is the dynamic viscosity

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of the hydraulic oil; l_p and d_p are the length and diameter of the hydraulic pipeline respectively.

2.3.4 Accumulator

The accumulator can store a certain amount of hydraulic oil, absorb pressure pulsation, buffer hydraulic shock, and improve the stability of the suspension system. According to the gas state equation, the pressure and volume of the accumulator can satisfy the following relationship:

$$P_{a0}V_{a0}{}^{\tau} = P_{at}V_{at}{}^{\tau} \tag{9}$$

where P_{a0} and V_{a0} are the initial pressure and volume of the accumulator respectively; P_{at} and V_{at} are the pressure and volume of the accumulator at time t respectively; τ is the gas index, τ=1.4.

The volume of hydraulic oil in the accumulator has the following relationship:

$$V_{at} - V_{a0} = \frac{b(D_w{}^2 - D_n{}^2)}{4r_b}\int \mathrm{v}(t)dt \tag{10}$$

where $\mathrm{v}(t)$ is the velocity of the pull arm at time t.

Therefore, the pressure of the accumulator can be calculated by the following equation:

$$P_{at} = \left(\frac{V_{a0}}{V_{a0} + \frac{b(D_w{}^2 - D_n{}^2)}{4r_b}\int \mathrm{v}(t)dt}\right)^{\tau} \tag{11}$$

2.3.5 Energy recovery module

The energy recovery module of the MEH regenerative suspension system is the hydraulic motor-generator. The hydraulic motor drives the generator to rotate and generate electricity, and the energy is recovered and stored in the battery through the energy recovery circuit to complete the energy recovery. For the convenience of analysis, the generator is decomposed into an ideal generator, internal resistance and internal inductance, and the energy recovery circuit is equivalent to a varistor.

The high-pressure hydraulic oil flowing out of the vane damper drives the hydraulic motor to rotate after rectification. The speed and torque of the hydraulic motor can be calculated from equation (12) and equation (13):

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$$\omega_m = \eta_{v2}\frac{2\pi Q_m}{V_m} \tag{12}$$

$$T_m = \eta_m\frac{\Delta P_m V_m}{2\pi} \tag{13}$$

where ω_m and T_m are the speed and torque of the hydraulic motor respectively; η_{v2} and η_m are the volumetric efficiency and mechanical efficiency of the hydraulic motor respectively; V_m is the displacement of the hydraulic motor; ΔP_{m} is pressure drop of the hydraulic motor.

The hydraulic motor is coaxially integrated with the generator, so the torque and speed of the generator are:

$$\omega_g = \omega_m \tag{14}$$

$$T_g = T_m \tag{15}$$

where ω_g and T_g are the torque and speed of the generator respectively.

The output voltage and input torque of the generator are calculated from the EMF equation:

$$U_e = k_e\omega_g \tag{16}$$

$$T_g = J_g\frac{d_\omega}{d_t} + k_t I \tag{17}$$

where U_e and I are the output voltage and output current of the generator respectively; k_e, k_t and J_g are the electromotive force constant, torque constant and moment of inertia of the generator respectively.

According to Kirchhoff's voltage law:

$$U_e - L_{in}\frac{d_I}{d_t} - I(R_{in} + R_{ex}) = 0 \tag{18}$$

where L_{in} and R_{in} are the internal inductance and resistance of the generator respectively, the internal inductance of the generator is generally negligible; R_{ex} is the external resistance of the generator.

The relationship of the pressure drop and the flow of the motor is:

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$$\Delta P_{\mathrm{m}} = \frac{4\pi^2 J_g \eta_{v2}}{V^2 \eta_m} \dot{Q}_{\mathrm{m}} + \frac{4\pi^2 k_t k_e \eta_{v2}}{V^2 \eta_m (R_{in} + R_{ex})} Q_{\mathrm{m}} \tag{19}$$

The inertia of the generator rotor can be regarded as the equivalent inertia together with the hydraulic motor. Therefore, the pressure drop of the hydraulic motor can be simplified as:

$$\Delta P_{\mathrm{m}} = \frac{4\pi^2 k_t k_e \eta_{v2}}{V^2 \eta_m (R_{in} + R_{ex})} Q_{\mathrm{m}} \tag{20}$$

2.3.6 Damping force

The equation for calculating the damping force of the vane damper is:

$$F_D = 2\int_{D_n/2}^{D_w/2} P_{vane} b \frac{S}{r_b} \mathrm{dS} = \frac{b}{4r_b}({D_w}^2 + {D_n}^2) P_{vane} \tag{21}$$

where F_D is the damping force; P_{vane} is the working pressure of the vane damper.

From the hydraulic circuit shown in Fig. 4, it can be known that:

$$P_{vane} = \begin{cases} \Delta P_v + \Delta P_p + \Delta P_m + P_{\mathrm{at}} & v > 0 \\ -(\Delta P_v + \Delta P_p + \Delta P_m + P_{\mathrm{at}}) & v < 0 \end{cases} \tag{22}$$

Therefore, the damping force of the MEH regenerative suspension system is:

$$F_D = \begin{cases} \frac{b}{4r_b}({D_w}^2 + {D_n}^2)(\Delta P_v + \Delta P_p + \Delta P_m + P_{\mathrm{at}}) & v > 0 \\ -\frac{b}{4r_b}({D_w}^2 + {D_n}^2)(\Delta P_v + \Delta P_p + \Delta P_m + P_{\mathrm{at}}) & v < 0 \end{cases} \tag{23}$$

2.3.7 Energy recovery power

After the above calculation, the output current I of the generator can be obtained by combining equations (15) and (17) as:

$$I = \frac{T_m}{ik_t} = \frac{\pi k_e \eta_{v1} \eta_{v2} b({D_w}^2 - {D_n}^2)}{2r_b V(R_{in} + R_{ex})} v \tag{24}$$

Therefore, the rational energy recovery power of the MEH regenerative suspension system is:

$$P_{re} = I^2 R_{ex} = \frac{\pi^2 {k_e}^2 {\eta_{v1}}^2 {\eta_{v2}}^2 b^2 ({D_w}^2 - {D_n}^2)^2}{4{r_b}^2 V^2 (R_{in} + R_{ex})^2} v^2 R_{ex} \tag{25}$$

2.4 Multiphysics coupling simulation model

The MEH regenerative suspension system has the conversion process of hydraulic energy to mechanical energy and then to electrical energy. The vibration reduction and

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energy recovery performance of the suspension system are affected by the coupling of multiphysics fields such as flow field, solid field, temperature field, and magnetic field. Therefore, using the power bond graph theory, a multiphysics coupling interface of the MEH regenerative suspension system simulation model is established in AMESim. The suspension actuator and the electrical control system model of the suspension system are built in AMESim according to the working principle of the suspension. The components in the Thermal hydraulic library are used to analyze the temperature change during the operation of the suspension. Fluent is used to analyze the flow field of the vane damper and hydraulic motor, and calculate the volumetric efficiency of the hydraulic components by calculating the flow resistance and considering the gap to calculate the leakage. Ansoft is used to analyze the magnetic field of the permanent magnet motor and calculate the reluctance. Through the cloud map, it can be seen that the magnetic field distribution of the generator is uniform, the magnetic flux leakage is small, and the magnetic field lines are uniformly distributed and relatively dense, which meets the design requirements. The fluid-solid-thermal-magnetic multiphysics coupling simulation model of the MEH regenerative suspension system is finally built as shown in Fig. 6.

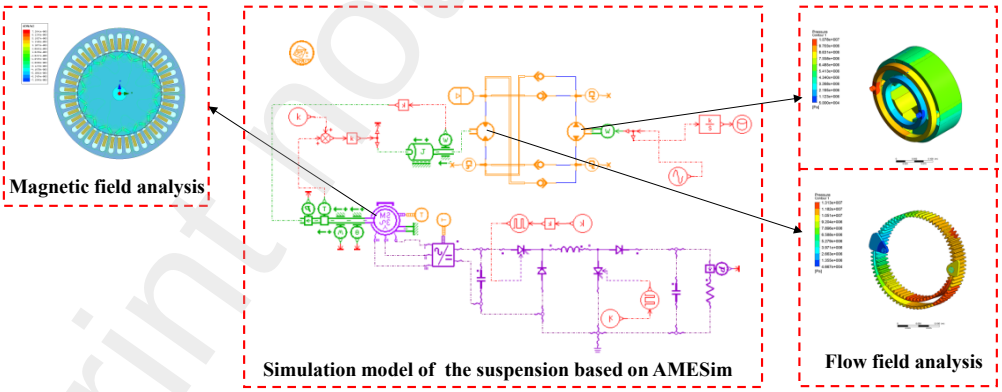

Fig.6. Multiphysics coupling simulation model of MEH regenerative suspension system

The main parameters in the simulation model of the MEH regenerative suspension system are set as shown in Table 1 according to the structural design size and the design parameters of the energy recovery circuit. The calculation methods of the equivalent moment of inertia and equivalent friction torque are referred to reference [30]. The volumetric efficiency of the hydraulic motor and vane damper has a great influence on

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the damping force and energy recovery power of the system and they are greatly affected by the actual environment during operation. The parameters set in the simulation model are only ideal values.

Table 1 Main parameters of MEH regenerative suspension system

Component	Parameter	Symbol	Value	Unit
Vane damper	Outer diameter	D_w	250	mm
	Inner diameter	D_n	112	mm
	Length of pull arm	r_b	252	mm
	Volumetric efficiency	η_{v1}	0.9	—
Hydraulic motor	Displacement	V	8	ml/r
	Volumetric efficiency	η_{v2}	0.80	—
	Mechanical efficiency	η_m	0.90	—
Accumulator	Volume	V_{a0}	0.8	L
	Initial pressure	P_{a0}	1.2	MPa
Pipeline	Length	l_p	40	mm
	Diameter	d_p	13	mm
Generator	Equivalent moment of inertia	J_g	0.00476	Kg·m^2
	Equivalent friction torque	T_f	5.346	N·m
	Internal resistance	R_{in}	0.13	Ω
Energy recovery Circuit	PWM frequency	f_p	10000	Hz
	Duty cycle	k	0.6	—
	Inductance	L	1000	μH
	Capacitance	C	260	μF
	Resistance	R_{ex}	Adjustable	Ω

3 Energy recovery module test

3.1 Test Design

In order to verify the conversion efficiency of hydraulic energy to mechanical energy and then to electrical energy and the actual power generation effect, the energy recovery module test is carried out. The energy recovery module consists of a hydraulic motor, a generator and an energy recovery circuit. The hydraulic motor is a four-quadrant internal meshing gear motor, and the generator is a 10kW BLDC motor. The test mechanism is shown in Fig.7. The vane damper is replaced by a hydraulic pump station to provide hydraulic energy to the energy recovery module. The accumulator is installed at the outlet of the hydraulic pump station to absorb pressure pulsation and

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prevent hydraulic shock. The hydraulic pump station adopts Parker's Compax3 servo hydraulic controller, which can output constant or sinusoidal flow as required. The hydraulic oil flows into the hydraulic motor after being rectified by the hydraulic commutator and drives the hydraulic motor to rotate to convert the hydraulic energy into mechanical energy, which in turn drives the generator to rotate and generate electricity.

The signal acquisition and data processing system are shown in Fig. 8. The flow signal and the inlet and outlet pressure signals of the hydraulic motor are collected through the flow sensor and the pressure sensor. The torque signal and speed signal output by the hydraulic motor are collected through the torque sensor and the speed sensor. The voltage signal and current signal are collected through the voltage sensor and the current sensor. The hydraulic energy, mechanical energy and electrical energy value and conversion efficiency of the energy recovery module can be obtained through data processing on the PC. The vane continuously swings when the vane damper is running, and the volume of hydraulic oil discharged from the vane damper is the volume swept by the vane. When the sinusoidal excitation frequency is 1.63Hz, and the excitation amplitude is 50mm (the swing angle amplitude is 11.44° at this time), the theoretical output flow of the vane damper (volume efficiency is 1) is 30L/min. By adjusting the servo controller of the hydraulic pump station, the amplitude of the constant flow and sinusoidal flow is 30L/min, and the performance of the energy recovery module is tested under this input flow.

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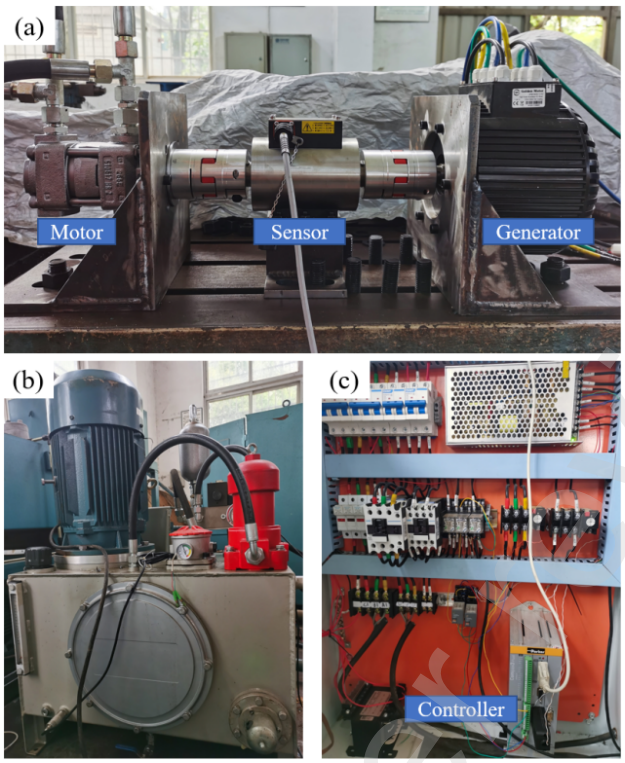

Fig. 7. Energy recovery module test device. (a) energy recovery module, (b) hydraulic pump station, (c) control system.

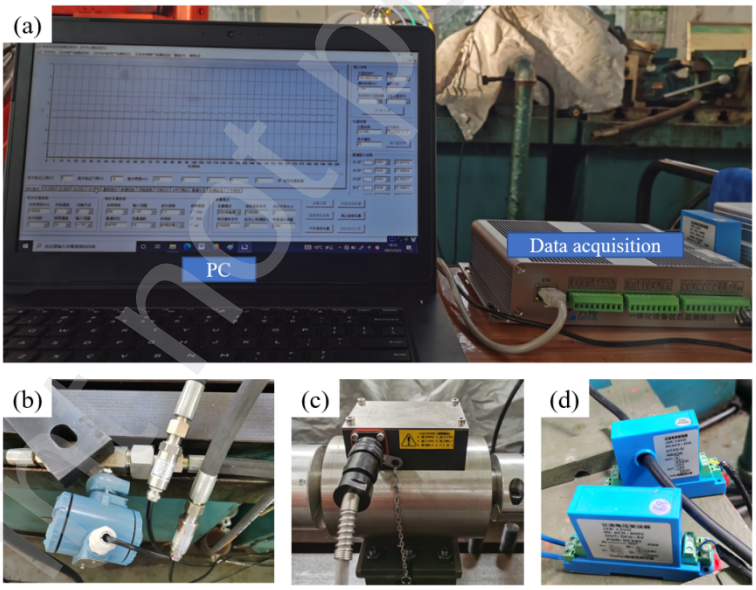

Fig. 8. Data collection and processing system. (a) data collection and processing, (b) hydraulic signal sensor, (c) mechanical signal sensor, (d) electrical signal sensor.

3.2 Test results

It can be seen from equations (21) and (22) that the output damping force of the EMH regenerative suspension system is mainly determined by the working pressure of the vane damper, and the working pressure of the vane damper is mainly determined

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by the pressure difference of the hydraulic motor. Therefore, the damping force of the EMH regenerative suspension system can be adjusted by changing the pressure difference of the hydraulic motor. In the energy recovery module, the power flow is from the hydraulic energy input of the hydraulic motor to the electrical energy output, ignoring the intermediate loss link, the power flow equation is:

$$P = \Delta P_{\mathrm{m}} Q_{\mathrm{m}} = \omega_{\mathrm{m}} T_{\mathrm{m}} = U_{\mathrm{g}} I_{\mathrm{g}} \tag{26}$$

It can be seen from equation (26) that the pressure difference of the hydraulic motor can be adjusted as long as the output current of the generator is changed, and the output damping force can be finally adjusted through the energy reverse control strategy. The output current of the generator is changed by adjusting the external resistance during the test. It can be seen from Fig. 9. that the pressure difference of the hydraulic motor decreases with the increase of the external resistance. Therefore, the pressure difference of the hydraulic motor can be adjusted by changing the external resistance of the energy recovery module, and finally achieve the semi-active control purpose.

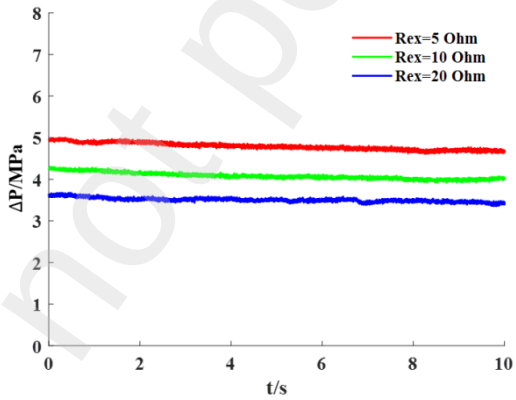

Fig. 9. Pressure difference of hydraulic motor

In order to accurately calculate the hydraulic energy-mechanical energy-electrical energy conversion efficiency of the energy recovery module, the constant flow condition is used for the test. Under the constant flow condition of 30L/min, the power value of the energy recovery module under different external load resistances are shown in Fig. 10, and the data is summarized in Table 2. When the external resistance is 5Ω, the generation power of the energy recovery module is 1104W, the hydraulic-electric conversion efficiency is 40.4%, and the mechanical-electric conversion efficiency is 83.6%. After being rectified by the hydraulic commutator, the energy recovery module

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can effectively realize the conversion and recovery of hydraulic-mechanical-electrical energy. However, the hydraulic-mechanical-electrical energy conversion efficiency of the energy recovery module decreases with the increase of the external resistance.

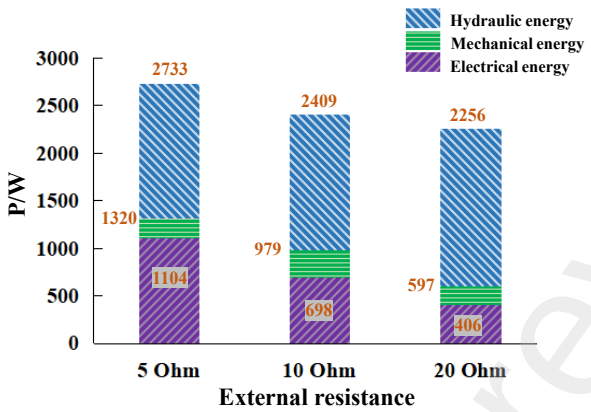

Fig. 10. Hydraulic energy-mechanical energy-electrical energy of energy recovery module

Table 2

Hydraulic-mechanical-electrical power and efficiency of energy recovery module under different external resistances

resistance (Ω)	Hydraulic power (W)	Mechanical power (W)	Electrical power (W)	H-E efficiency (%)	M-E efficiency (%)
5	2733	1320	1104	40.4%	83.6%
10	2409	979	698	29.0%	71.9%
20	2256	597	406	18.0%	70.1%

The generated power of the energy recovery module under sinusoidal input hydraulic frequencies of 0.1Hz, 0.5Hz and 1Hz are shown in Fig. 11-Fig. 13. Under the sinusoidal condition, the maximum generating power of the energy recovery module decreases with the increase of the external resistance, which is consistent with the constant flow condition. However, due to the long hydraulic pipeline between the hydraulic pump station and the hydraulic commutator in the test device, a part of the energy will be offset subject to the response speed of the hydraulic system during rectification. Therefore, the maximum generated power decreases as the sinusoidal frequency increases. The test results show that the length of the hydraulic pipeline will seriously affect the energy recovery effect of the system. The MEH regenerative suspension system adopts an integral structure design, and the hydraulic commutator is arranged on the diaphragm of the vane damper, which shortens the hydraulic pipeline to the maximum extent. This structure can effectively avoid hydraulic loss during

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rectification and improve energy recovery efficiency.

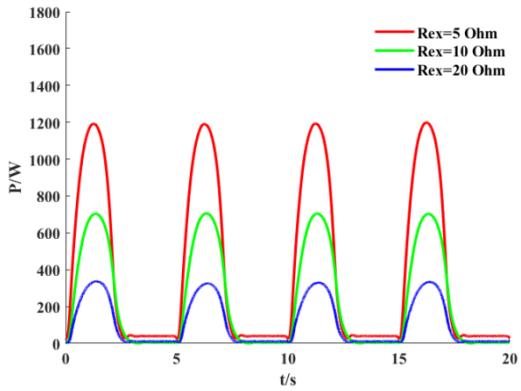

Fig. 11. The power generated by the energy recovery module at 0.1Hz

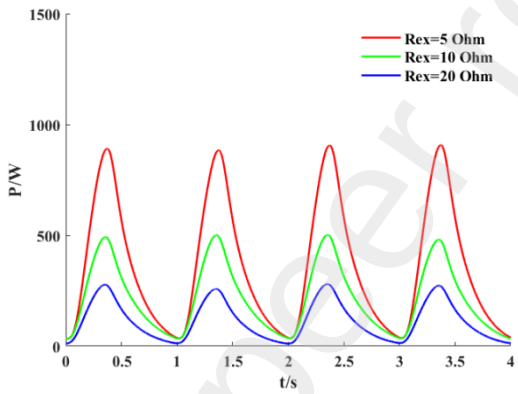

Fig. 12. The power generated by the energy recovery module at 0.5Hz

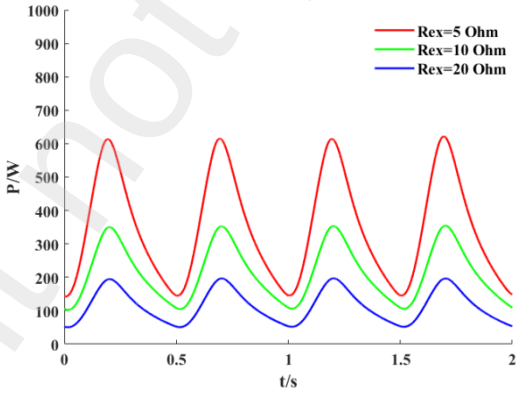

Fig. 13. The power generated by the energy recovery module at 1Hz

4 Characteristics analysis of MEH regenerative suspension system

The parameters of the multiphysics coupling simulation model are set according to the hydraulic motor efficiency measured by the test, and the damping and energy recovery characteristics of the MEH regenerative suspension system are simulated and analyzed.

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4.1 Analysis of Damping Characteristics

The novel MEH regenerative suspension system is a special suspension that can realize energy recovery. Its primary function is to provide damping force to meet the vibration reduction requirements of high-speed tracked vehicle. The amplitude of the excitation amplitude can be represented by the swing amplitude of the vane damper, and the relationship is shown in the follow equation:

$$\theta = \arcsin(S/r_b) \tag{27}$$

where θ is the vane swing angle amplitude, S is the excitation amplitude, and r_b is the length of the pull arm.

When the sinusoidal excitation frequency is 1.67Hz and the external load resistance is 10Ω, the damping characteristics of the suspension under different swing angle amplitudes (±5°, ±10°, ±15°, ±20°, ±25°, the swing angle is positive when the vane swings clockwise, and the swing angle is negative when the vane swings counterclockwise) of the vane damper are shown in Fig. 14 and Table 3. When the vane swing angle amplitude is 5°, the excitation amplitude is 23mm, the excitation speed is 0.227m/s, and the maximum damping force of the system is 7.92kN. When the vane swing angle amplitude is 25°, the excitation amplitude is 107mm, the excitation speed is 1.153m/s, and the maximum damping force of the system is 25.63kN. Under the passive condition, the damping force increases with the increase of the excitation amplitude, and the absolute value of the damping force is equal when the vane swings clockwise or counterclockwise, which is the same as equation (23). The maximum output damping force density of the MEH regenerative suspension system can reach 1349kN/m^3.

Table 3

Damping characteristics of the suspension under different swing angle amplitudes

Angle amplitude /°	Excitation amplitude /mm	Excitation speed /(m/s)	Maximum damping force /kN
5	23	0.227	7.92
10	44	0.459	14.88
15	65	0.691	20.08
20	86	0.922	23.55
25	107	1.153	25.63

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Under the sinusoidal excitation with the frequency of 1.67Hz and the amplitude of 50mm (the swing angle amplitude of the vane damper is 11.44°), when the external resistance increases from 5Ω to 25Ω, the damping characteristic of the suspension is shown in Fig. 15. The damping force of the suspension decreases with the increase of the external resistance. When the external resistance increases from 5Ω to 25Ω, the maximum damping force of the system decreases from 25.12kN to 9.38kN, and the adjustment factor of the maximum damping force is 2.68 times. The suspension system can control the damping force by adjusting the external resistance to achieve semi-active control.

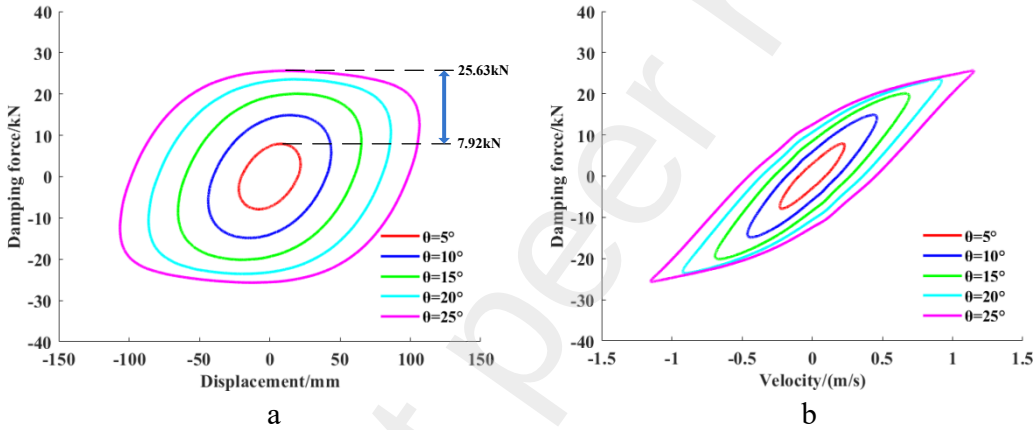

Fig.14. Damping characteristics of the suspension under different swing angle amplitudes. (a)*F*-*s* curves, (b)*F*-*v* curves

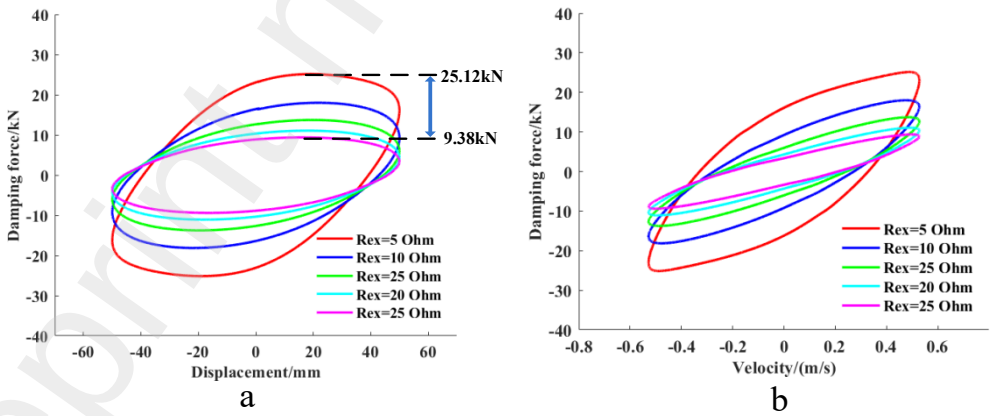

Fig. 15. Damping characteristics of the suspension under different external resistances. (a)*F*-*s* curves, (b)*F*-*v* curves

4.2 Energy recovery characteristics of the suspension under sinusoidal excitation

The effect of sinusoidal excitation frequency and amplitude on the energy recovery

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power of the suspension is shown in Fig. 16. The energy recovery power of the MEH regenerative suspension system under the excitation frequency of 1 Hz with different swing angle amplitudes are shown in Fig. 16(a). The energy recovery power increases significantly with the increase of the swing angle amplitude of the vane damper. The average energy recovery power of the system is shown in Fig. 16(b) and Table 3. When the swing angle amplitude of the vane damper increases from 5° to 25° at 1Hz frequency, the average energy recovery power of the system increases from 273W to 3483W.When the excitation frequency increases from 0.5Hz to 2.5Hz at 10°amplitude, the average energy recovery power of the suspension increases from 202W to 3371W. The average energy recovery power of the suspension has a high positive correlation with the excitation amplitude and excitation frequency, and the MEH regenerative suspension system has a better energy recovery effect under large amplitude conditions.

Adjusting the external resistance can not only change the damping characteristics of the system but also affect the energy recovery characteristics of the system. When the sinusoidal excitation frequency is 1.67 Hz and the excitation amplitude is 50 mm, the energy recovery characteristics of the system under different external resistances are shown in Fig. 17. The average energy recovery efficiency of the system in one cycle can be calculated according to equation (28):

$$\eta = \frac{\int_0^T P_{re} dt}{\int_0^T |T_{\mathrm{d}} \omega| dt} \tag{28}$$

where P_{re} is the energy recovery power; T_{d} is the output torque of the vane damper; ω is the speed of the vane damper; T is the full stroke motion period.

The energy recovery power and energy recovery efficiency of the suspension decrease significantly with the increase of the external resistance. When the external resistance increases from 5Ω to 25Ω, the average energy recovery power in one cycle is reduced from 2972W to 979W, and the average energy recovery efficiency of the suspension in one cycle is about 40%. The MEH regenerative suspension system can theoretically achieve kilowatt level energy recovery power with high efficiency.

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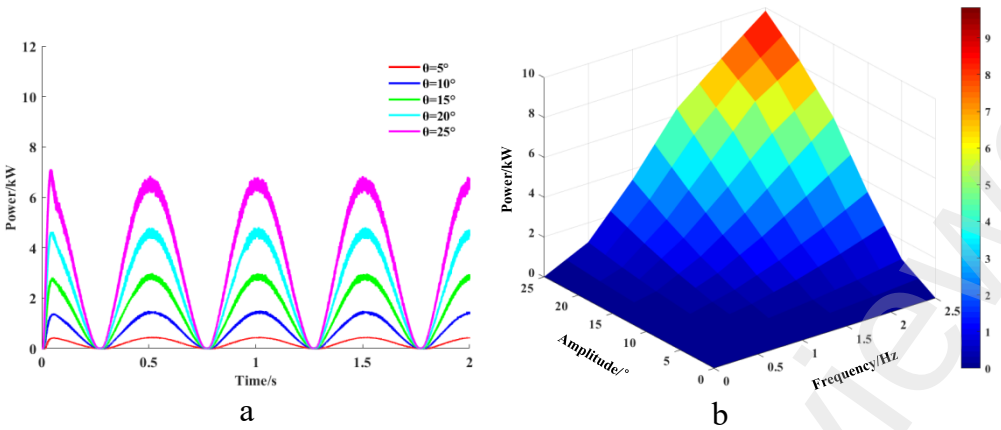

Fig. 16. Influence of sinusoidal excitation characteristics on energy recovery power. (a) energy recovery power at 1Hz, (b) influence of excitation frequency and amplitude on the average energy recovery power.

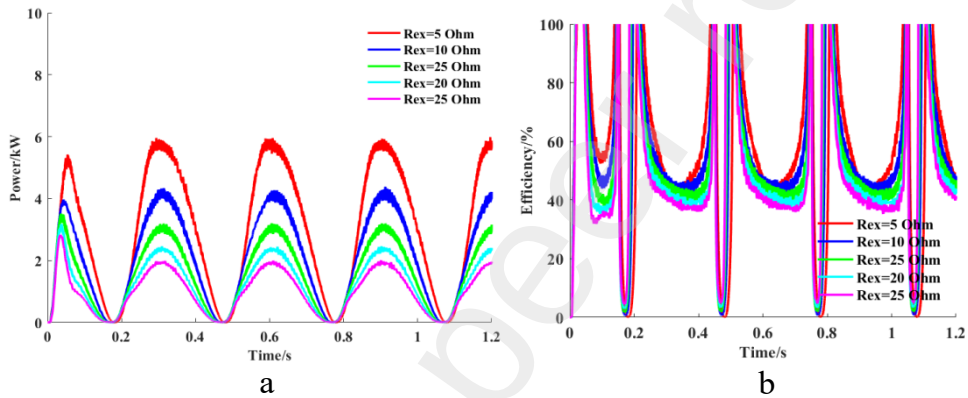

Fig. 17. Influence of external resistance on energy recovery characteristics. (a) energy recovery power, (b) Energy recovery efficiency.

4.3 Energy recovery characteristics under random road excitation

The driving conditions of high-speed tracked vehicle is complex and changeable. Typical random road excitations are used to analyze the energy recovery characteristics of the MEH regenerative suspension system under different road grades and different driving speeds. In the process of establishing the random road excitation model, the power spectral density (PSD) is used to describe the statistical characteristics of the road surface roughness, and the Gauss white noise method is used to establish the road surface roughness excitation time-domain model, as shown in equation (29):

$$\dot{q}(t) = -2\pi n_{00} u q(t) + 2\pi n_0 \sqrt{G_q(n_0) u} W(t) \tag{29}$$

where $q(t)$ is the random road elevation displacement; n_0 is the reference spatial frequency, n_0=0.1$_{\mathrm{m}}{}^{-1}$; n_{00} is the lower cut-off spatial frequency, n_{00}=0.011$_{\mathrm{m}}{}^{-1}$;

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$G_q(n_0)$ is the road roughness coefficient; $W(t)$ is Gauss white noise with a mean value of 0; u is the vehicle speed.

Considering the actual driving speed of the high-speed tracked vehicle and the off-road conditions, the energy recovery efficiency is studied under the road grades B, C, D, and E at the vehicle speeds of 5m/s, 10m/s, and 20m/s, respectively. As shown in Fig. 18(a), the energy recovery power of the suspension increases with the increase of road grade, and the fluctuation range becomes larger. As shown in Fig. 18(b) and Table 4, the average energy recovery power of the MEH regenerative suspension system increases significantly with the increase of road grade and vehicle speed. When driving on E-class road at a speed of 20m/s can achieve an average energy recovery power of 5293W. At this time, the energy recovery power density reaches 278.5 kW/m^3, and the MEH regenerative suspension system can achieve high power density energy recovery.

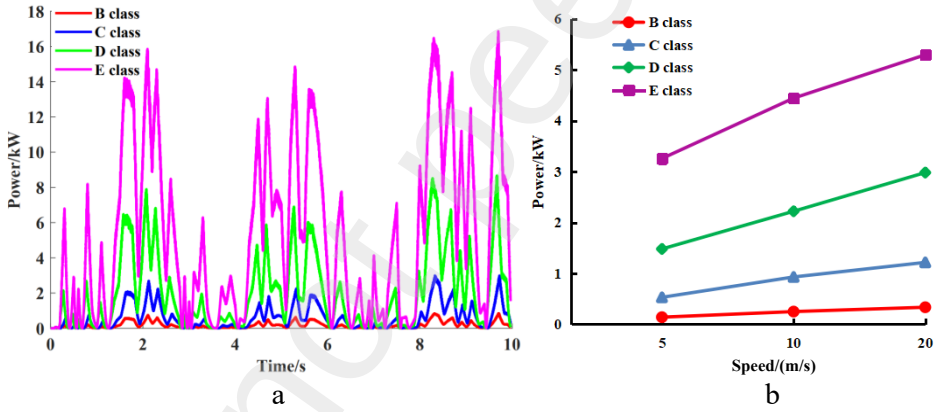

Fig. 18. Energy recovery power. (a)Energy recovery power under different road grades of 20 m/s; (b) Influence of vehicle speed and road grade on the average energy recovery power.

Table 4

Average energy recovery power under different speeds and road grades(W).

Road grade	Speed/(m/s)		
	5	10	20
B class	142	251	335
C class	534	937	1216
D class	1480	2180	2984
E class	3256	4442	5293

5 Conclusion

In this paper, a novel high-power density MEH regenerative suspension system

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suitable for high-speed tracked vehicle is proposed. Firstly, the working principle of the MEH regenerative suspension system and the energy transfer and dissipation paths are introduced in detail. On this basis, the mathematical model of the suspension system and the multiphysics coupling simulation model are established. Then the energy recovery module test is designed to verify its energy recovery feasibility. The main conclusions include:

(1) The transmission of vibration energy in the suspension system goes through the conversion process of hydraulic energy, mechanical energy and electrical energy, and there are many dissipation paths in the process of vibration energy transmission and conversion.

(2) The test proves that the pressure difference of the hydraulic motor can be changed by changing the external resistance. And it can be seen from the simulation that the MEH regenerative suspension system can realize the damping force adjustment under passive and semi-active conditions through an energy reverse control strategy.

(3) It can be seen from the test that the energy recovery module can reach 40.4% of the hydraulic-electric energy conversion efficiency under the external resistance of 5Ω. The energy recovery power and efficiency decrease with the increase of the external resistance. And the length of the hydraulic pipeline has a great influence on the energy recovery effect of the suspension.

(4) The MEH regenerative suspension system can achieve kilowatt-level energy recovery power with an energy recovery efficiency of about 40% under ideal low-frequency conditions. The energy recovery power increases significantly with the increase in vehicle speed and road roughness. Driving on the E-class off-road road at a speed of 20m/s can reach an average energy recovery power of 5.293kW. The maximum energy recovery power density of the MEH regenerative suspension system reaches 278.5kW/m^3, which has an excellent energy recovery characteristic.

The simulation results show that the MEH regenerative suspension system can theoretically achieve kilowatt-level energy recovery power with high efficiency, but the energy recovery efficiency obtained from the test results is lower than the simulation value. Therefore, it is a key research topic in the future to carry out structural

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optimization design to reduce the loss of vibration energy transfer process, and improve the energy recovery efficiency of the MEH regenerative suspension system.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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